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Whole slide image redundancy reduction for efficient pathological diagnosis

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ABSTRACT

The rapid advancement of Computational Pathology (CP) has revolutionized pathological diagnosis and driven the development of intelligent diagnostic technologies. However, a major challenge in applying these technologies is processing Whole Slide Images (WSIs) with resolutions that can exceeding tens of millions of pixels (e.g., 10,000 × 10,000). While existing methods predominantly focus on WSI representation and discrimination, they often overlook efficiency—an increasingly critical concern given the growing demand for high-throughput diagnosis. This highlights the urgent need for efficient solutions. Notably, the recurrence of biological features and patterns in pathological images indicates substantial redundancy within WSIs, offering an opportunity for optimization. In this paper, we propose **WSIR^2**, a WSI Redundancy Reduction-based diagnostic framework. WSIR^2 incorporates a novel token compression module that strategically selects the top-k most informative patches and merges less informative ones, thereby reducing both the number of tokens and their feature dimensions. Additionally, it employs the cross-attention mechanism with $\mathcal{O}(n)$ complexity to construct a lightweight classifier for efficient classification. These innovations lead to a dramatic reduction in computational cost – achieving up to 24× lower GFLOPs compared to leading MIL methods – while maintaining diagnostic performance on par with state-of-the-art models. Experimental results demonstrate that WSIR^2 consistently delivers faster training and higher inference throughput, offering a compelling balance between accuracy and efficiency for large-scale pathological diagnostic tasks.

1. Introduction

Whole Slide Images (WSIs) scanned by digital slide scanners have profoundly changed the diagnostic paradigm of clinical pathology [1,2]. The laborious operation of targeting biological patterns under the classical microscope has been replaced by the ability to access the patients' pathological status via electronic devices, which has greatly streamlined pathologists' workflow. Concomitantly, the digitalization of pathology has introduced the challenge of efficiently and effectively processing the colossal volumes of ever-increasing WSIs, whose gigapixel-level resolution further aggravates the computational burden. In the field of computational pathology, numerous outstanding methods have been proposed to aid pathological diagnosis [3–6]. It is worth noting that although these methods have gained impressive performance on many general tasks and scenarios [7–10], few of them have explored the efficiency of WSI analysis. In the face of the growing demand for pathological diagnosis, there is still an urgent need to efficiently analyze WSIs to accelerate the associated diagnostic tasks and both qualitative and quantitative biological analyses.

Due to the gigapixels of WSI and scarce pixel-level annotations, previous works have assumed that WSI contains abundant information and have developed various effective methods within the framework of weakly-supervised learning (WSL) for pathological diagnosis, where only slide-level labels are available, while patch-level labels are not. Furthermore, based on multiple instance learning (MIL), which treats a WSI as a bag and patches from it as instances, numerous methods have constructed promising deep models to capture the relationship between bags and their labels, achieving impressive accuracy on general tasks [11–16]. However, most of these works overlook the practical demand for efficiency in clinical tasks. Although several methods have explored the efficiency of WSI diagnosis [17–19], few focus on optimizing computational cost and reducing redundant information within WSIs. Motivated by the ever-increasing volume of pathological images, it is crucial to improve the efficiency of MIL-based models.

On the other hand, compared to natural images, pathological images contain monotonous, similar biological patterns such as cells, tissues,

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and sub-organ structures. As Fig. 1 illustrates, patches from pathological images exhibit higher similarity with each other than patches from natural images, which indicates that reducing redundant patches can improve diagnostic efficiency while retaining critical information within a single image. Additionally, during the diagnostic process, pathologists focus on disease-related patterns or features against a regular background, *e.g.* a partial tumor region in the tissue landscape. This also suggests that only a few significant patches adequately represent the entire status of patients. All these cues provide evidence that excessive redundancy exists in WSI for pathological diagnosis. Thus, reducing redundancy in WSI could effectively tackle the limitations of efficiently processing gigapixel WSIs and satisfy the increasing demands of computer-aided diagnosis.

Motivated by the observations above, we propose a **WSI r**edundancy **r**eduction method, *i.e.* **WSIR**^{2}, for efficient WSI analysis and diagnosis. WSIR^2 is characterized by two modules: token compression and a lightweight classifier. In the token compression module, we simultaneously prune and merge tokens within a single block. Based on the efficient cross-attention mechanism with $\mathcal{O}(n)$ computational complexity and its interpretation of attention values as indicators of token significance, WSIR^2 achieves redundancy reduction of WSIs. It simultaneously reduces features along both the feature dimension and the token dimension by selecting *top-k* tokens and fusing the non-significant tokens. Then, a lightweight classifier is proposed to further reduce computational costs of features aggregation for WSI classification, which is constructed with several cross-attention blocks. Again, benefiting from the $\mathcal{O}(n)$ computational complexity of the blocks, its computational cost is further reduced.

To evaluate the efficiency and effectiveness of WSIR^2, we conducted extensive experiments on multiple datasets, including Camelyon16, TCGA-NSCLC, and UniToPatho. The experimental results demonstrate that WSIR^2 significantly reduces FLOPs and model size to 1.02 or 0.29 GFLOPs and 225K parameters on these datasets, while maintaining competitive performance compared to other MIL methods. Furthermore, incorporating the token compression module into other MIL methods also significantly accelerates their inference, demonstrating its broad applicability. These results also highlight that redundancy within WSIs has been comprehensively neglected in previous works, and the efficiency of WSI tasks should be seriously considered to meet the increasing demands of computer-aided diagnosis. WSIR^2 establishes an efficient paradigm for efficient WSI diagnosis, which has built the foundation for the further development.

Our contributions are concluded as follows:

- We propose WSIR^2, an efficient MIL method for pathological diagnosis and analysis. It is composed of a token compression module and a lightweight classifier.
- We propose token compression, which utilizes cross-attention to select *top-k* patches and merge the non-*top-k* patches, thereby accelerating inference efficiency.
- Extensive experimental results have validated the effectiveness and efficiency of WSIR^2, demonstrating its potential in resolving the bottleneck of high-throughput diagnoses.

2. Related work

2.1. Multiple instance learning for WSI classification

MIL-based pathological diagnosis utilizes readily available slide-level diagnostic results as labels for model training and has gained widespread attention in the computational pathology community. Early MIL-based methods are rooted in an instance-based approach, in which a model is first trained for patch-level prediction and then produces the slide-level prediction by aggregating the patch-level predicted logits using mean or max pooling [20,21]. As this scheme introduces erroneous pseudo-labels for patches and fails to capture robust relationships between contextual information and slide-level predictions, it has gradually been superseded by embedding-based methods. The embedding-based approach takes patch-level embeddings as input and directly predicts the slide-level label. Because patch-level embeddings, rather than only their labels, are available, this branch of methods generally achieves better results. For example, MIL-RNN [11] treats a WSI as a sequence of patches and accounts for patch permutation variation. CLAM [9] constructs an attention-based model to perform instance-level clustering over representative regions, which are identified to constrain and refine slide-level features. TransMIL [15] models relationships among patches via self-attention (SA) [22]. HIPT [23] builds a contrastive learning framework with hierarchical features for multi-scale fusion of WSIs. HAG-MIL [24] proposes a hierarchical gated attention mechanism to leverage multi-scale features of WSIs. To address the data deficiency of pathological images, DTFD-MIL [19] generates various sub-bags from WSIs for data augmentation. Remix [17] generates slide-level samples based on clustering prototypes. MHIM [25] employs a siamese structure with a consistency constraint to explore potential hard instances. IBMIL [26] achieves de-confounded bag-level prediction based on an interventional training framework. In this work, we specifically explore redundancy reduction within these mechanisms and introduce cross-attention with learnable queries. This design enhances scalability and generalization, enabling our method to integrate seamlessly into a variety of attention-based architectures while preserving efficiency.

2.2. Redundancy reduction for vision tasks

Redundancy reduction was first proposed in the context of neuroscience [27]. It has been instrumental in explaining the organization of the visual system and has led to numerous algorithms for supervised and unsupervised learning [28–31]. In vision tasks, this principle has been applied to vision model pre-training and efficient model construction [32–34]. For example, Rao et al. [35] employ sparse attention in vision transformers and retrain a network to dynamically filter tokens; Yin et al. [36] propose an adaptive token reduction mechanism to speed up inference without modifying the network architecture; Liang et al. [37] leverage cross-attention scores between the class token and other tokens to reorganize image tokens; and Bolya et al. [38] propose a general and lightweight matching algorithm to gradually combine similar tokens within a ViT. Nevertheless, these studies primarily focus on subject-specific natural images with fixed sizes. In this paper, applying redundancy reduction to gigapixel WSIs is rarely explored, and the prevalence of monotonous biological patterns within WSIs motivates our WSIR^2.

3. Methodology

3.1. MIL formulation

Firstly, we provide a brief explanation of embedding-based MIL. Taking binary classification as an example, to predict a target value Y_i of the ith bag, given a bag containing n instances $\{x_{i,1}, x_{i,2}, \dots, x_{i,n}\}$ while the instance-level labels $\{y_{i,1}, y_{i,2}, \dots, y_{i,n}\}$ are unknown, a binary classification can be defined as:

$$\mathrm{Y_i} = \begin{cases} 0 & \forall j, y_{i,j} = 0, \\ 1 & \exists j, y_{i,j} = 1, \end{cases} \tag{1}$$

where $y_{i,j} \in \{0,1\}$ and $j = 1, 2, \dots, n$.

Furthermore, the embedding-based MIL framework adopts a transformation function f to obtain instance representations and a MIL classifier S to predict the label of the bag based on these features, which can be expressed as:

$$\hat{\mathrm{Y}}_\mathrm{i} = \mathrm{S}\big(f(x_{i,1}), f(x_{i,2}), \dots, f(x_{i,n})\big). \tag{2}$$

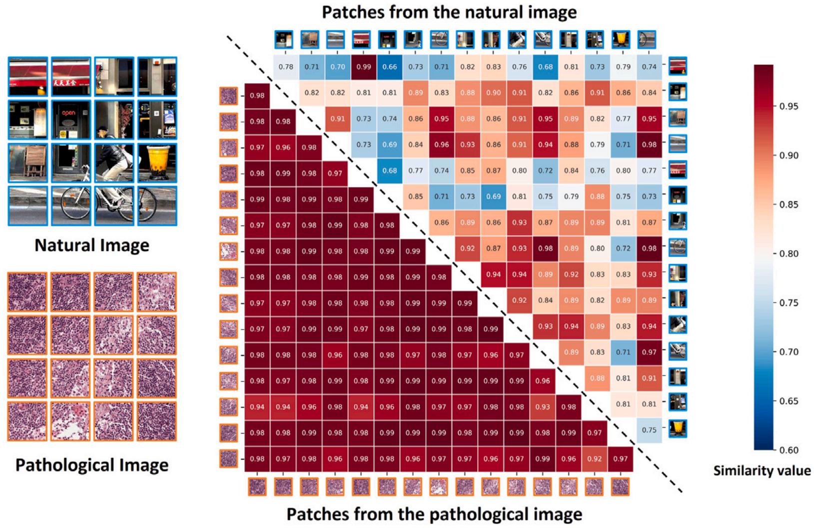

Fig. 1. Cosine-similarity of patches from natural image and pathological image. A general vision encoder (ViT-small, pre-trained with DINO on diverse data containing pathological images) is used to separately extract features from patches of the natural image and the pathological image as shown. Patches (4 × 4) from the pathological image have much larger similarity values with each other compared to natural image.

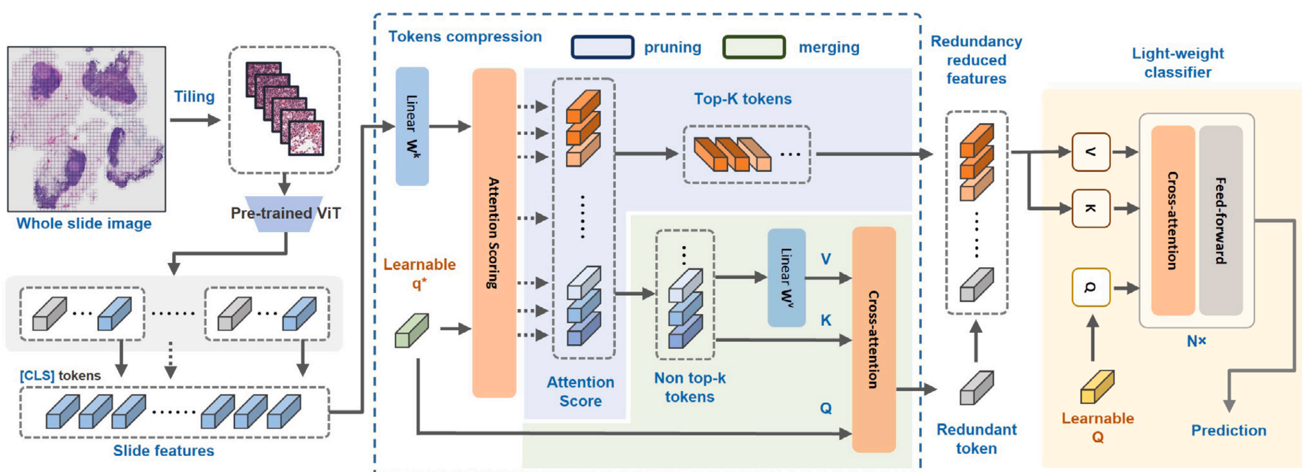

Fig. 2. Framework of WSIR[2]. Patches tiled from 20× magnification of the WSI are used for self-supervised learning. Pre-trained ViT-small is subsequently used to extract patch features at optimal magnification and then concatenated as slide-level features. WSIR[2] reduces the redundancy at patch-level and dimension-level, *i.e.*, reduces numbers and dimensions of patches. The MIL classifier could be any model including ours and others proposed.

3.2. WSI redundancy reduction

The frequent recurrence of similar biological patterns in WSIs suggests that disease-related semantics are globally sparse and largely redundant. Selecting informative patches for diagnosis offers an efficient paradigm, allowing MIL aggregation to focus on fewer yet more critical tokens. In this context, token compression is an effective means of mitigating WSI redundancy. However, unlike token pruning and merging in ViTs for natural images [30,38], token compression in WSIs cannot be performed progressively across multiple self-attention blocks, as the enormous number of tokens renders the direct application of a vanilla ViT impractical.

To overcome the limitations, we proposed WSIR[2] to achieve a balance between efficiency and accuracy. As illustrated in Fig. 2, the extracted patch-level features are concatenated into a slide-level feature, $\mathbf{X} \in \mathbb{R}^{n \times d}$, where n represents the number of patches and d is feature dimension. WSIR[2] performs features compression at the two dimensions: token pruning and merging to reduce the patch number n, and dimensionality reduction to reduce the feature dimension d. Concretely, patch-level features are first transformed to $\mathbf{K} \in \mathbb{R}^{n \times d'}$, which represents the compressed features that retain the most critical information for each patch. To reduce the number of patches that need to be aggregated, a learnable parameter $\mathbf{q}^* \in \mathbb{R}^{d'}$ is introduced to select the informative patches and merge these non-significant patches simultaneously via cross-attention mechanism. Specifically, $\mathbf{q}^*$ not only assigns attention weights a_i to each token in $\mathbf{K}$ but also compresses redundant tokens into a single representative token. The computation of attention values is formulated as follows:

$$\mathbf{K} = \mathbf{W}^k \mathbf{X} = \{\mathbf{k}_i \mid i = 0, 1 \ldots, n\} \tag{3}$$

$$\mathbf{A} = \mathrm{softmax}\left(\frac{\mathbf{q}^* \mathbf{K}^\top}{\sqrt{d'}}\right) = \{a_i \mid i = 1, 2, \ldots, n\} \tag{4}$$

where $\mathbf{W}^k$ is a learnable projection, and d' is the shared dimensionality of $\mathbf{q}^*$ and $\mathbf{k}_i$.

Then, the *top-k* tokens with the highest attention values are selected from $\mathbf{K}$ according to a predefined ratio r (*i.e.*, the *top-k* ratio). The *top-k* tokens are concatenated as the informative tokens $\mathbf{K}^t \in \mathbb{R}^{k \times d'}$, while other tokens are less informative tokens, $\mathbf{K}^r \in \mathbb{R}^{(n-k)\times d'}$:

$$k = \max(1,\ \lceil n \cdot r \rceil), \tag{5}$$

$$\mathbf{K}^t = \{\mathbf{k}_i \mid i \in \mathrm{TopK}(a_i,\ k)\}, \tag{6}$$

$$\mathbf{K}^r = \{\mathbf{k}_i \mid i \notin \mathrm{TopK}(a_i,\ k)\} \tag{7}$$

where $\lceil\cdot\rceil$ denotes the ceiling function and k is the number of retained tokens.

For the less informative tokens, we compress them into a single redundant token $\mathbf{x}_r \in \mathbb{R}^{d'}$, which aims to preserve some background information of the WSI:

$$\mathbf{V}^r = \mathbf{W}^v\mathbf{K}^r, \tag{8}$$

$$\mathbf{A}^r = \{a_i \mid i \notin \mathrm{TopK}(a_i,\ k)\}, \tag{9}$$

$$\mathbf{x}^r = \mathbf{A}^r\mathbf{V}^r, \tag{10}$$

where $\mathbf{V}^r \in \mathbb{R}^{(n-k)\times d'}$ is the linearly projected representation of $\mathbf{K}^r$, sharing the same dimensionality d'. To minimize computational cost, we reuse the attention scores of $\mathbf{q}^*$ to $\mathbf{K}^r$ (*i.e.,* $\mathbf{A}^r$) in Eq. (4). And a linear mapping is directly applied to $\mathbf{K}^r$ to obtain $\mathbf{V}^r$, where $\mathbf{W}^v \in \mathbb{R}^{d'\times d'}$ introduces significantly less computational burden compared to applying it to the raw features $\mathbf{K}$. This simple yet effective operation serves as an approximation and is further optimized during training.

Subsequently, the pruned tokens ($\mathbf{K}^t$) and the merged token ($\mathbf{x}^r$) are concatenated and fed into a lightweight classifier for prediction. Again, we employ the cross-attention mechanism as the basic building block of the classifier. Concretely, a class token $\mathbf{Q}$ is initialized as the query, which attends to the input features and compresses them into a single global representation $\mathbf{X}^g$. The predicted label $\hat{Y}$ is then obtained from this representation as follows:

$$\mathbf{X}^k = \mathrm{concat}(\mathbf{K}^t,\ \mathbf{x}^r), \tag{11}$$

$$\mathbf{X}^g = \mathrm{softmax}\left(\frac{\mathbf{Q}(\mathbf{W}^{k'}\mathbf{X}^k)^\top}{\sqrt{d'}}\right)(\mathbf{W}^{v'}\mathbf{X}^k), \tag{12}$$

$$\hat{\mathbf{Y}} = \mathrm{S}(\mathbf{X}^g) \tag{13}$$

where $\mathrm{S}(\cdot)$ is a scoring function that maps the global representation to the final prediction, and $\mathbf{W}^{k'}$ and $\mathbf{W}^{v'}$ are learnable projection matrices applied to $\mathbf{X}^k$. In summary, WSIR^2 leverages token compression in conjunction with a lightweight classifier, substantially reducing computational cost and thereby enabling efficient diagnosis.

4. Experiments

4.1. Datasets

WSIR^2 is extensively evaluated on three public datasets: Camelyon16 [39], TCGA-NSCLC, and UniToPatho [40]. Camelyon16 is collected for metastasis detection in breast cancer and contains two classes (Normal/Tumor). It officially provides 270 training samples and 129 testing samples, which can be tiled into roughly 57.18 million patches at 20× magnification. TCGA-NSCLC includes two subtypes of lung cancer: Lung Adenocarcinoma (LUAD) and Lung Squamous Cell Carcinoma (LUSC). We obtained 1042 slides from https://www.cancer.gov/ccg/access-data. All slides of TCGA can be tiled into 145.63 million patches at 20× magnification. We further split this dataset into a training set and a test set using an 8:2 ratio, and evaluate all methods on the test set. UniToPatho is a public dataset comprising 292 WSIs, of which we utilized 243 from https://ieee-dataport.org/open-access/unitopatho. All slides of UniToPatho can be tiled into 9.87 million patches at 20× magnification. This dataset includes four disease subtypes (NORM, HP, HG, LG), and we follow the official train/test split.

Table 1
Classification accuracy and AUC comparison on Camelyon16, TCGA-NSCLC, and UniToPatho datasets. Results are averaged over five independent runs. The highest performance is shown in **bold**, and the second best is <u>underlined</u>. r denotes the *top-k ratio* used in WSIR^2.

Datasets		Camelyon16		TCGA-NSCLC		UniToPatho	
Methods		Acc.	AUC	Acc.	AUC	Acc.	AUC
Mean		0.6279	0.5904	0.8804	0.9460	0.6847	0.8353
Max		0.8992	0.9491	0.9398	0.9830	0.6937	0.8390
ABMIL		0.9302	0.9656	0.9450	0.9812	0.7849	0.8968
DSMIL		0.9205	0.9526	0.9408	0.9808	0.7685	0.9009
TransMIL		0.9238	0.9542	0.9450	0.9830	0.6904	0.8215
CLAM-SB		0.9256	0.9545	0.9453	<u>0.9831</u>	0.7838	0.9156
CLAM-MB		0.9302	0.9536	<u>0.9458</u>	**0.9832**	0.7827	0.9071
DTFD-MIL		**0.9390**	<u>0.9643</u>	0.9446	0.9736	0.7973	<u>0.9239</u>
MHIM-MIL		0.9264	0.9570	0.9366	0.9785	0.7854	0.8996
WSIR^2	$r = 1.00$	<u>0.9349</u>	**0.9674**	**0.9466**	0.9810	**0.8108**	**0.9273**
	$r = 0.50$	0.9315	0.9654	0.9446	0.9813	<u>0.8074</u>	0.9243
	$r = 0.10$	0.9202	0.9429	0.9394	0.9819	0.7854	0.9132
	$r = 0.01$	0.8815	0.9132	0.9387	0.9825	0.7838	0.9072

4.2. Implementation details

We construct the datasets by tiling the WSIs into non-overlapping 224 × 224 patches at a magnification of 20×, discarding background patches. We then conduct self-supervised pre-training with DINO [41] on the datasets. In the pre-training step, the backbone is ViT-small/16 (12 transformer blocks with a hidden dimension of 384, patch size of 16 × 16). For the pre-training setup, we follow the original training settings and only adjust the batch size due to limited hardware resources. After pre-training, features extracted by ViT-small are used in subsequent WSI classification. For a fair comparison of multiple MIL methods, the averaged accuracy and AUC over 8 random repetitions are reported. In the classification training step, cross-entropy loss is adopted. The AdamW [42] optimizer with an initial learning rate of 2×10^{-4}, and a cosine annealing scheme without warm restarts [43] for learning rate scheduling, is used to update the model weights. The main hyperparameter of WSIR^2 is *topk_ratio*, abbreviated as r, which is separately set to $[0.01, 0.1, 0.5, 1.0]$ to evaluate the redundancy in WSIs. To evaluate model efficiency, we report the model parameter counts and FLOPs. The FLOPs differ across datasets, and we report the counts for processing 14,000 and 4000 patches separately—14,000 being the average number of patches per WSI from Camelyon16 and TCGA-NSCLC, and 4000 from UniToPatho. All experiments are conducted with 8× NVIDIA GeForce RTX 3090 GPUs. The unit of model parameter counts is K.

4.3. Main results

We summarize the main comparison of classification performance and computational efficiency in Tables 1–3. WSIR^2 consistently demonstrates a favorable balance between diagnostic accuracy and efficiency across three benchmark datasets (Camelyon16, TCGA-NSCLC, and UniToPatho).

Classification Performance. As shown in Table 1, WSIR^2 achieves competitive accuracy and AUC on all datasets. With a lightweight classifier and token dimension $d' = 96$, and without pruning ($r = 1.00$), WSIR^2 attains the competitive accuracy on Camelyon16 (0.9349), TCGA-NSCLC (0.9466), and UniToPatho (0.8108). These results indicate that WSIR^2 can preserve critical diagnostic information while using substantially fewer model parameters.

Efficiency and Trade-off Analysis. Table 2 compares WSIR^2 with representative MIL methods in terms of model size, computational cost, and inference speed. WSIR^2 maintains the lowest parameter count (225K) among all methods. In terms of GFLOPs, even without pruning ($r = 1.00$), WSIR^2 requires only 1.36 G for 14,000 patches, which is 3–25× lower than other MIL methods. Correspondingly, inference

Table 2
Efficiency comparison of WSIR^2 and other mainstream MIL methods. Model parameters (Params) and floating-point operations (GFLOPs) are reported in thousands (K) and billions (G), respectively. The left and right GFLOPs correspond to processing 14,000 patches (average in Camelyon16 and TCGA-NSCLC) and 4000 patches (UniToPatho), respectively. Inference speed (Infer. Speed), training time per epoch (Train Time), and memory usage (Mem. Usage) are measured on TCGA-NSCLC.

Methods		Params	GFLOPs	Infer. Speed	Train time	Mem. Usage
ABMIL		300	4.24, 1.20	965 WSIs/s	0.75 s/epoch	0.50 GB
DSMIL		298	4.24, 1.21	342 WSIs/s	1.04 s/epoch	0.49 GB
TransMIL		2344	34.61, 10.27	62 WSIs/s	10.60 s/epoch	2.3 GB
CLAM-MB		461	6.58, 1.86	574 WSIs/s	1.10 s/epoch	0.57 GB
DTFD-MIL		459	4.69, 1.30	419 WSIs/s	2.50 s/epoch	1.04 GB
MHIM-MIL		526	15.01, 4.25	173 WSIs/s	1.76 s/epoch	0.57 GB
WSIR^2	r = 1.00	225	1.36, 0.39	985 WSIs/s	1.06 s/epoch	0.43 GB
	r = 0.50	225	1.02, 0.29	1001 WSIs/s	1.04 s/epoch	0.43 GB
	r = 0.10	225	0.75, 0.21	1046 WSIs/s	0.98 s/epoch	0.43 GB
	r = 0.01	225	0.69, 0.20	1096 WSIs/s	0.95 s/epoch	0.43 GB

Table 3
Efficiency and performance comparison when integrating WSIR^2 into other MIL methods. The results are evaluated on TCGA-NSCLC. The units of model parameters and FLOPs are in thousands (K) and billions (G), respectively. r denotes the *top-k ratio*. Accuracy (ACC.) and area under the ROC curve (AUC) reflect diagnostic performance, while GFLOPs, inference speed, training time, and memory usage assess computational efficiency.

Methods	*top-k ratio*	ACC.	AUC	Param.	GFLOPs	Infer. Speed	Train time	Mem. Usage
ABMIL	w/o WSIR^2	0.9450	0.9812	300	4.24	965 WSIs/s	0.75 s/epoch	0.50 GB
	$r = 0.50$	0.9402	0.9743	75	0.75	1257 WSIs/s	0.92 s/epoch	0.43 GB
	$r = 0.10$	0.9366	0.9700	75	0.70	1372 WSIs/s	0.85 s/epoch	0.43 GB
	$r = 0.01$	0.9091	0.9584	75	0.68	1451 WSIs/s	0.84 s/epoch	0.43 GB
DSMIL	w/o WSIR^2	0.9408	0.9808	298	4.24	342 WSIs/s	1.04 s/epoch	0.49 GB
	$r = 0.50$	0.9386	0.9746	75	0.75	406 WSIs/s	1.13 s/epoch	0.44 GB
	$r = 0.10$	0.9322	0.9738	75	0.69	426 WSIs/s	1.08 s/epoch	0.44 GB
	$r = 0.01$	0.9234	0.9674	75	0.68	450 WSIs/s	1.02 s/epoch	0.43 GB
TransMIL	w/o WSIR^2	0.9450	0.9830	2344	34.61	62 WSIs/s	10.60 s/epoch	2.30 GB
	$r = 0.50$	0.9372	0.9795	2252	17.14	136 WSIs/s	8.71 s/epoch	1.38 GB
	$r = 0.10$	0.9221	0.9695	2252	4.08	153 WSIs/s	8.19 s/epoch	0.69 GB
	$r = 0.01$	0.9091	0.9627	2252	1.24	163 WSIs/s	8.02 s/epoch	0.57 GB

Table 4
Efficiency of WSIR^2 under varying token dimensions (d') and *top-k* ratios (r). d' controls the local feature dimension, and r controls the proportion of preserved tokens.

Hyperparamter		Param ↓	FLOPs ↓
d'	r	[K]	[G]
96	1.00	225	1.36,0.39
	0.50		1.02,0.29
	0.10		0.75,0.21
	0.01		0.69,0.20
192	1.00	818	4.25,1.21
	0.50		2.94,0.83
	0.10		1.87,0.53
	0.01		1.63,0.46
384	1.00	3110	14.78,4.19
	0.50		9.57,2.71
	0.10		5.31,1.51
	0.01		4.36,1.23

speed reaches 985 WSIs/s, significantly higher than other MIL methods. Memory usage is also reduced (0.43 GB vs. 0.49–2.3 GB for other models), demonstrating the lightweight nature of WSIR^2. When applying token pruning ($r = 0.50, 0.10, 0.01$), GFLOPs decrease further to 1.02 G, 0.75 G, and 0.69 G, respectively, while inference speed increases to 1001–1096 WSIs/s. Accuracy remains competitive across datasets (Table 1). These results indicate that WSIR^2 can achieve substantial computational savings and faster inference than existing MIL models, while offering flexible trade-offs between efficiency and performance.

Integration with Existing MIL Models. We further evaluated WSIR^2 as a slide-level feature compression module for representative MIL models, including ABMIL, DSMIL, and TransMIL (Table 3). Incorporating WSIR^2 substantially reduces computational cost and accelerates inference across all architectures. For instance, with ABMIL at $r = 0.50$, GFLOPs drop from 4.24 G to 0.75 G and throughput increases from 965 to 1257 WSIs/s, while accuracy only slightly decreases (0.9450 → 0.9402). TransMIL achieves similar gains, with throughput more than doubling at $r = 0.50$. These results highlight the generalizability of WSIR^2 for reducing redundancy in WSIs and improving the efficiency of diverse MIL methods without severely compromising predictive performance.

Overall, WSIR^2 provides a practical solution for high-throughput WSI analysis. It maintains competitive accuracy, drastically reduces computational cost and memory usage, and accelerates inference. These advantages make it particularly suitable for clinical deployment, where efficiency and speed are critical for timely diagnosis.

4.4. Ablation

We further investigate the effects of varying token dimensions on token pruning and merging. We report the performance of WSIR^2 with $d' = 96$ in Table 1, and compare it to $d' = 196$ and $d' = 384$ in Fig. 3. The efficiency metric is also reported in Table 4. For WSIs with abundant tokens, such as TCGA-NSCLC, employing higher token dimensions requires a greater number of tokens to construct complete WSI representations for effective classification. Conversely, fewer feature dimensions ($d' = 96$) exhibit stable robustness as tokens are pruned.

In UniToPatho, which contains fewer tokens on average (approximately 4000) and sparse biological tissue-structure information, deploying a larger model with more hidden dimensions introduces noise to the MIL model. This often leads to severe overfitting and degraded accuracy on the test set—a trend also observed in TransMIL (see Table 1). In contrast, WSIR^2 with $d' = 96$ effectively suppresses noise and achieves optimal performance. These observations validate the presence of redundancy within WSIs and emphasize the robustness of WSIR^2 under varying token dimensionalities.

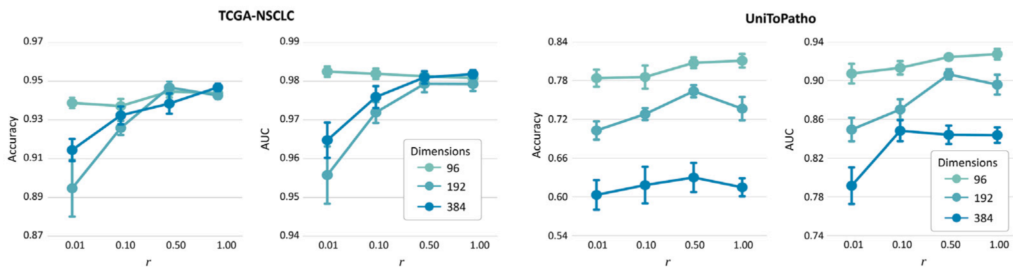

Fig. 3. Effect of token dimension (d') and *top-k ratio* (r) on WSIR^2 performance. Accuracy and AUC are evaluated on TCGA-NSCLC and UniToPatho datasets to illustrate the trade-off between computational efficiency and predictive performance. Experiments are conducted with $d' \in \{96, 192, 384\}$ and $r \in \{0.01, 0.10, 0.50, 1.00\}$.

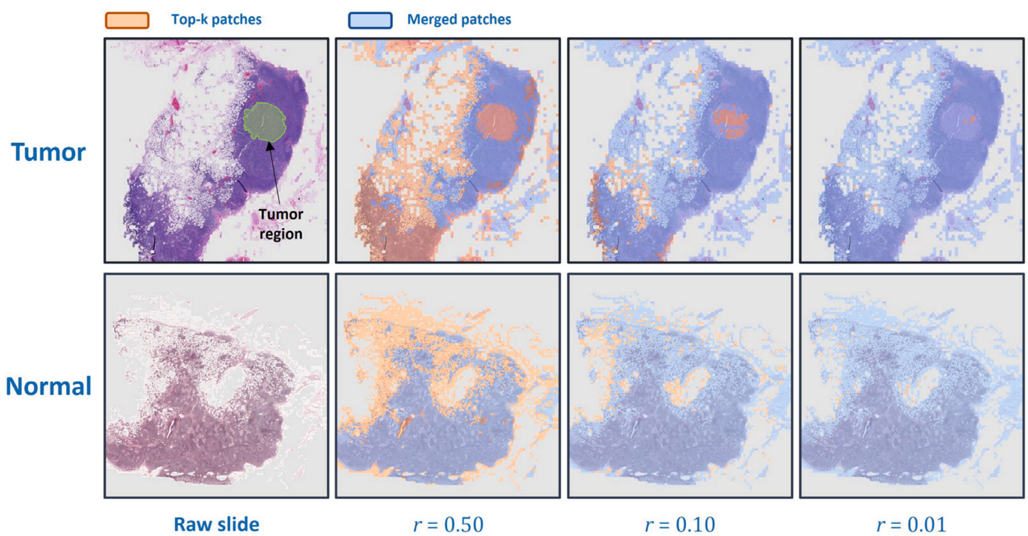

Fig. 4. Visualization of pruned and merged patches in WSIs using WSIR^2. Two representative Camelyon16 test samples are shown, illustrating the effect of WSIR^2 under varying *top-k ratios*.

4.5. Visualization

To examine the behavior of pruned and merged patches in WSIR^2, we visualize two Camelyon16 WSIs using *top-k* ratios of {0.01, 0.10, 0.50}. As shown in Fig. 4, patches with highly similar biological morphology, such as parenchymal cells (background information), are merged. WSIR^2 consequently focuses more on indispensable diagnostic regions, such as fat cells and tumor cells within the tissue.

5. Discussion

WSIR^2 effectively addresses redundancy in WSIs by leveraging token pruning and merging alongside a lightweight classifier, achieving substantial reductions in computational cost and model size while preserving critical diagnostic information. Its plug-and-play design enables acceleration of various MIL models, demonstrating strong generalizability, and the visualization results confirm its focus on diagnostically relevant regions. However, WSIR^2 relies on a fixed *top-k ratio*, which may be suboptimal for highly heterogeneous WSIs, and extreme pruning can slightly degrade accuracy. Additionally, it depends on pre-extracted patch-level features and does not explicitly capture local spatial context, which may limit performance for tasks requiring fine-grained detail. Future improvements could include adaptive token selection, multi-scale or hierarchical attention to better model local and global context, integration with end-to-end trainable backbones to reduce preprocessing overhead, and extension to other pathological tasks, thereby enhancing both efficiency and clinical applicability.

6. Conclusion

In this paper, we propose WSIR^2, an efficient framework for WSI diagnosis. WSIR^2 introduces a token compression module leveraging the cross-attention mechanism and a lightweight classifier. The concept of diagnostic acceleration is incorporated throughout the design of WSIR^2. Extensive experiments on multiple datasets demonstrate that WSIR^2 achieves highly competitive classification performance while maintaining remarkable efficiency. Extending WSIR^2 to a broader range of WSI analysis tasks warrants further investigation. Therefore, our future work will focus on whole end-to-end pipeline of WSI diagnosis and facilitating additional downstream tasks.

CRediT authorship contribution statement

Shijie Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Zhineng Chen:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Feng-Jung Chen:** Supervision, Resources, Formal analysis, Data curation. **Xieping Gao:** Supervision, Project administration, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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